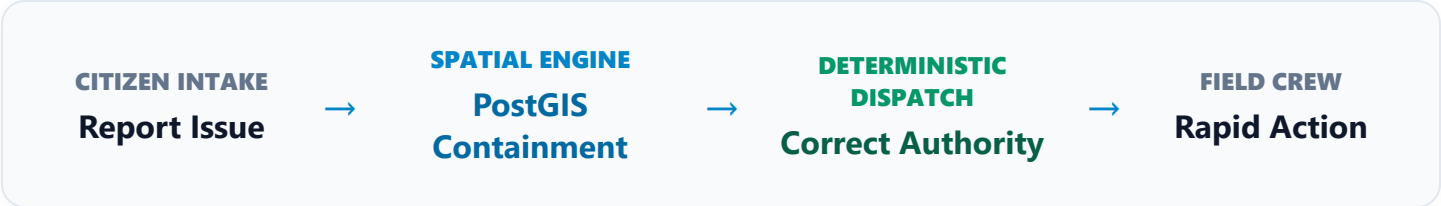


GOVERNMENT OF KARNATAKA • CIVIC INTELLIGENCE ENGINE

ADAPTIVE CIVIC ROUTING INTELLIGENCE SYSTEM

“Report once. Route correctly. Resolve with accountability.”

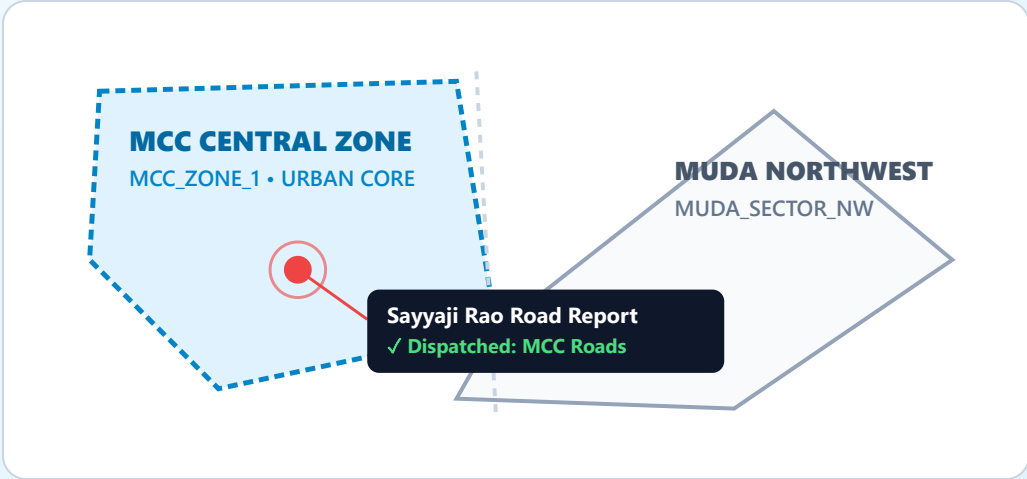


MYSURU URBAN REGION (MCC / MUDA)

ACTIVE CADASTRE: MYS_2026_V2

Automated Municipal Routing Pipeline

Deterministic spatial containment replacing manual administrative guesswork.



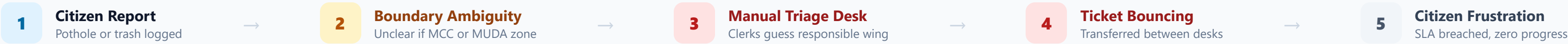
Spatial Reference: EPSG:4326 (WGS 84)

Deterministic Spatial Routing

Why Civic Complaints Get Stuck

A citizen submits an issue once, but administrative boundary ambiguity delays resolution for weeks.

STATUS QUO WORKFLOW FAILURE



01

Boundary Ambiguity

Administrative boundaries do not align with citizens' colloquial street knowledge or postal pincodes.

02

Manual Triage

Intake officers spend hours manually re-reading tickets to guess whether an issue belongs to MCC or MUDA.

03

Routing Errors

A complaint assigned to the wrong zone gets rejected and re-entered, resetting SLA countdowns without accountability.

04

Lost Provenance

When municipal wards are redrawn or renamed, past decisions lose legal validity and cannot be audited.



THE FUNDAMENTAL HACKATHON QUESTION

“How do we determine the correct authority from location — reliably and reproducibly?”

Deterministic Spatial Computing

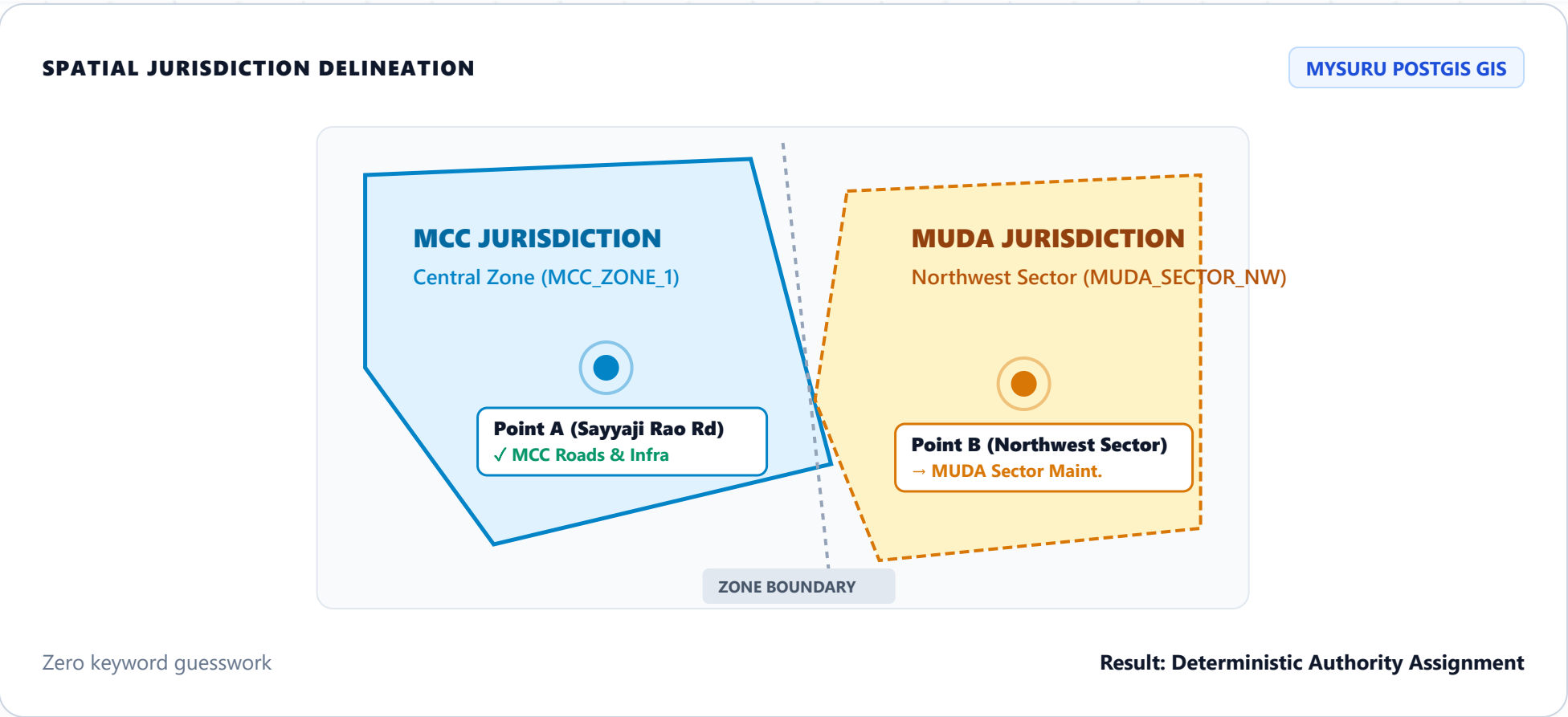
Location Determines Responsibility

The same civic issue type requires completely different routing depending on where it falls on the map.

Civic routing cannot be solved with text search or keyword tags alone.

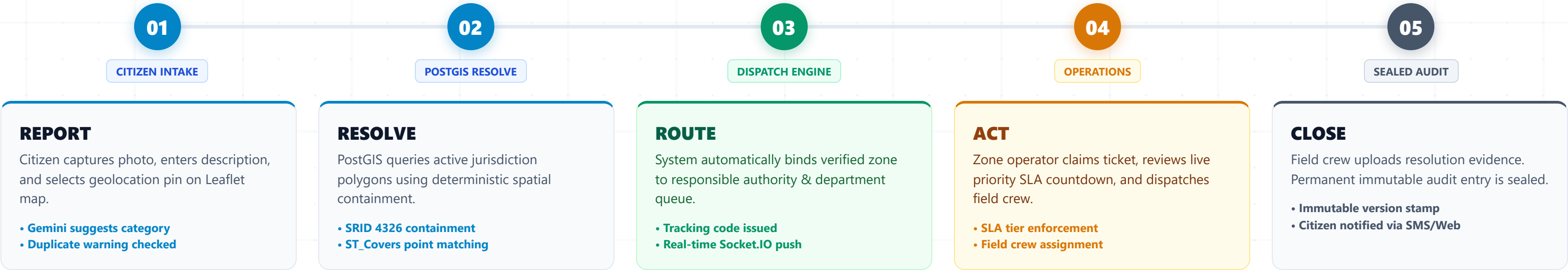
A pothole on Sayyaji Rao Road falls inside MCC Central Zone and routes to MCC Roads & Infrastructure. A road issue across the sector boundary routes to MUDA Sector Maintenance.

01. CITIZEN COORDINATE
Point (Latitude, Longitude) in EPSG:4326
02. ACTIVE JURISDICTION BOUNDARY
MultiPolygon containment match (ST_Covers / ST_Contains)
03. COMPETENT CIVIC AUTHORITY
Mysuru City Corporation (MCC) vs MUDA Development Authority
04. OPERATIONAL QUEUE & CREW
MCC Roads & Infrastructure (MCC_ROADS)



From Citizen Report to Responsible Field Team

A seamless 5-stage pipeline delivering automated spatial routing and complete audit provenance.



END-TO-END AUTOMATION ADVANTAGE

Zero manual triage delay. Real-time ticket progression from citizen report to verified closure in a single audited system.

Deterministic Logic

Audit Verified

AI Assists. GIS Decides.

Clear architectural boundaries: Generative AI cannot hallucinate or override civic jurisdiction.

ASSISTIVE INTELLIGENCE

GEMINI AI

Classification Assistance

Natural language understanding to assist citizens in structuring their complaints without bureaucratic friction.

- **Smart Category Tagging:**
Tags issue as "Pothole", "Garbage", or "Streetlight" (`^AI_SUGGESTED``).
- **Citizen UX Acceleration:**
Eliminates manual scrolling through 40+ civic subcategories.

STRICT ARCHITECTURAL GUARDRAIL
Gemini AI is NEVER permitted to decide ward jurisdiction, select authorities, or dispatch field crews.

VS

STRICT
SEPARATION

DETERMINISTIC ENGINE

POSTGIS (SRID 4326)

Spatial Containment & Routing

Mathematical point-in-polygon resolution providing verifiable authority mapping without hallucination risk.

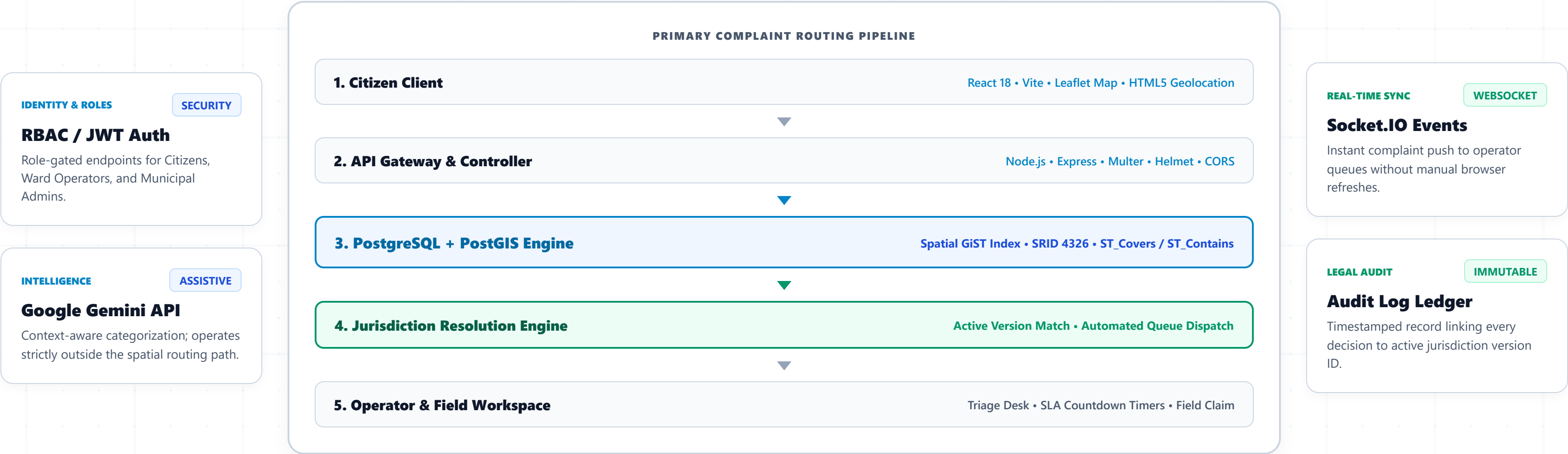
- **Cadastral Containment:**
Uses `ST_Covers(boundary_geom, point)` across active ward polygons.
- **Deterministic Authority Mapping:**
Binds polygon results directly to official civic department routing rules.

MUNICIPAL GOVERNANCE GUARANTEE
Deterministic jurisdiction resolution: Every routing decision is reproducible and permanently auditable.

“AI can assist the interpretation. The map decides the authority.”

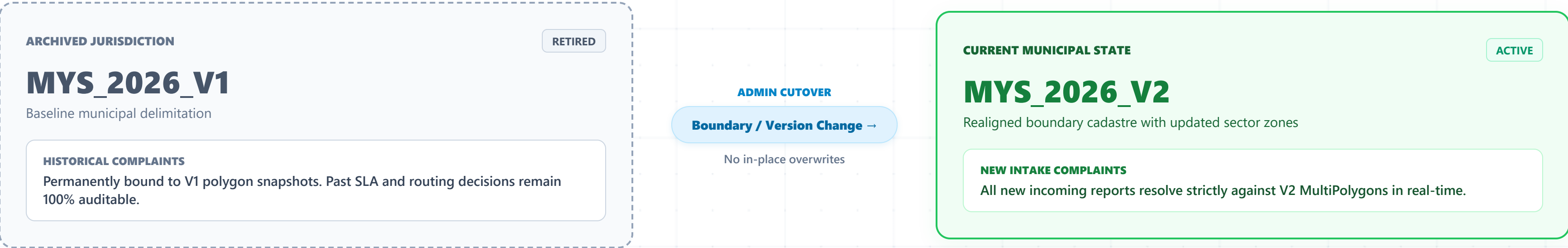
How the System Works Underneath

A modular, production-tested architecture separating user intake, spatial computation, and dispatch.



Municipal Boundaries Change. History Cannot.

How immutable jurisdiction versioning preserves complete legal auditability during ward redistricting.



01 / WHY NOT OVERWRITE GEOMETRIES?

In-place table updates corrupt historical records. If a boundary shifted in August, a July inquiry cannot verify why a complaint went to Central Zone.

02 / AUDIT PROVENANCE GUARANTEE

Every complaint record stores foreign key `jurisdiction_version_id` alongside geometry coordinate point.

03 / PROVEN IN PRODUCTION

Verified in test suite: Version cutover transactions activate new cadastre without downtime or corrupting historical decisions.

Built, Deployed, and Verified

Genuine application execution proof: Sayyaji Rao Road complaint routed via deterministic PostGIS logic.

GENUINE COMPLAINT PHOTO

LIVE INTAKE EVIDENCE



CANONICAL COMPLAINT CODE:
HM-CIV-2026-000656

VERIFIED LOCATION:
Sayyaji Rao Rd (MCC Zone 1)

CATEGORY DETECTED:
POTHOLE (AI_SUGGESTED)

ACTIVE CADASTRE:
MYS_2026_V2 (ACTIVE)

VERIFIED EXECUTION LOG & AUDIT CHAIN

01 Citizen Submission Intake REST API: POST /api/complaints

HTTP 201

02 Spatial Coordinate Point Construction PostGIS: ST_SetSRID(ST_MakePoint(lng, lat), 4326)

SRID 4326

03 Jurisdiction Containment Resolved Bound to MCC Central Zone 1 via ST_Covers

MATCHED

04 Department Queue Assignment Routed: Roads & Infrastructure (MCC_ROADS)

DISPATCHED

05 Operator Workspace Sync Pushed live via Socket.IO; SLA timer initiated

SLA TRACKED

06 Immutable Audit Ledger Record Logged with timestamp, version hash, and actor ID

SEALED

LIVE DEPLOYED MVP:

<https://adaptive-civic-routing.vercel.app>

Designed with Real-World Constraints in Mind

Demonstrating architectural maturity: Transparent analysis of current limitations and scaling roadmaps.

01 / SPATIAL BOTTLENECK

COMPUTE

GIS Compute at Scale

CURRENT LIMITATION

Evaluating complex MultiPolygons across multiple zones introduces computational overhead under concurrent report bursts.

Implemented in MVP: Spatial GiST indexing on polygon boundaries and point geometries with spatial containment.

FUTURE SCALING ROADMAP

Pre-computed quadkey spatial tiling and Redis caching of active zone centroids.

02 / FIELD CONNECTIVITY

NETWORK

Offline Field Operations

CURRENT LIMITATION

Field workers inspecting subterranean drainages or dense markets frequently encounter mobile data blackouts.

Implemented in MVP: Graceful server degradation, network disconnection status banners, and reconnect polling.

FUTURE SCALING ROADMAP

Service Worker PWA with IndexedDB store-and-forward queue for complete offline logging.

03 / DATA VOLUME

DATABASE

Database Scaling Ceiling

CURRENT LIMITATION

A single PostgreSQL instance encounters write saturation during city-wide monsoon crisis complaint surges.

Implemented in MVP: `pg.Pool` connection management, normalized schema, and indexed foreign keys.

FUTURE SCALING ROADMAP

Read replicas for analytics, time-series audit partitioning, and zone-based shards.

“We built an honest, high-performance working system with clear architectural clarity on what comes next.”

Architectural Integrity

From One Ward to City-Wide Civic Intelligence

Scaling Mysuru's municipal governance with spatial precision and transparent public trust.

STRATEGIC DEPLOYMENT PHASES

01

Phase 1: Ward-Level Live Pilot

Deploy in MCC Central Commercial Zone with local sanitation and road engineering field crews.

02

Phase 2: Full Municipal GIS Integration

Expand to all MCC municipal zones, MUDA peripheral layouts, and inter-agency utility routing.

03

Phase 3: Kannada Voice & Text Interaction

Voice-to-text complaint logging in Kannada for grassroots accessibility via WhatsApp & IVR channels.

AI DISCLOSURE & RESPONSIBLE GOVERNANCE

Google Gemini API: Used strictly for classification assistance & category suggestions.

Antigravity AI: Used for rapid development pair-programming and verification tooling.

✓ **Immutable Guarantee:** All municipal boundaries, jurisdiction containment, and department routing decisions remain 100% deterministic PostGIS spatial queries.

Live MVP: <https://adaptive-civic-routing.vercel.app>

GitHub: github.com/kkaribasappakl-eng/adaptive-civic-routing

Team Spark: HM26-E9C5 • Hemanth N, Karibasappa KL, Pooja M, Soumya Doddamani

OUR CIVIC COMMITMENT

“Report once. Route correctly. Resolve with accountability.”

HackMysuru 1.0 Submission